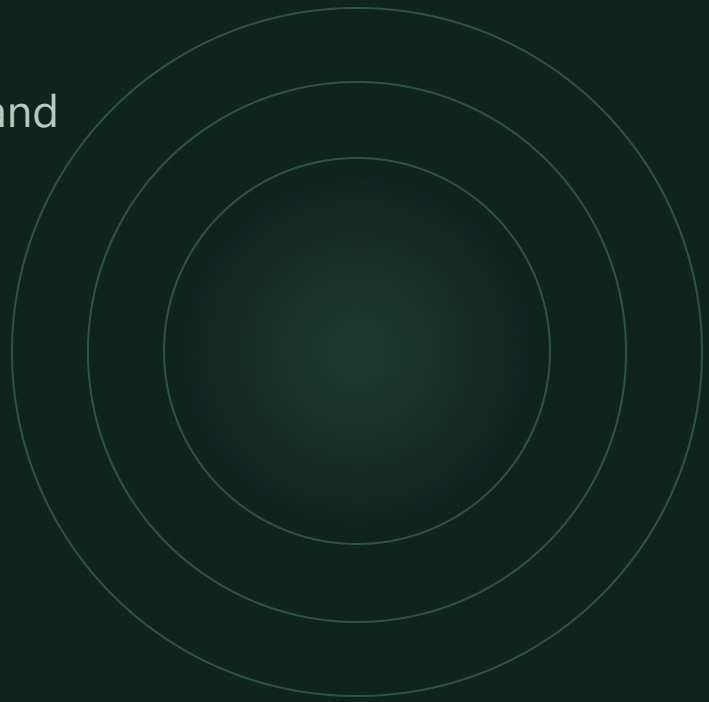
MEDIBOT

PROJECT PRESENTATION · OCTOBER 2026

Clearer medicine information. Safer conversations.

A safety-first conversational prototype for catalogue lookup, price snapshots, and sourced general health information.



Bun · TypeScript · SQLite · Groq · Tavily

01 · THE IDEA

Useful information, with clear boundaries

People often need to find a product in a large catalogue, understand what data is actually available, and know where an answer came from. MediBot brings those steps into a conversational interface.

FIND

Catalogue lookup

Search imported product names and pack sizes from a local SQLite catalogue.

COMPARE

Price snapshots

Show source-provided values and label their origin —not as a live pharmacy offer.

EXPLAIN

Sourced answers

Optional medical-reference web search returns clickable source links.

Product principle: an empty or incomplete dataset is not a reason to invent facts. Say what is missing, preserve provenance, and keep medical decisions with qualified professionals.

02 · REQUEST JOURNEY

From a question to a bounded answer

1 · Ask
User acknowledges the medical disclaimer and sends a question.

›

2 · Guard
Server checks known restricted terms and high-signal indirect phrasing.

›

3 · Retrieve
Groq can call guarded SQLite tools; Tavily is optional for general references.

›

4 · Check
Generated text is checked for restricted names and diagnosis language.

›

5 · Cite
Web-based answers display HTTPS source links and a general-info notice.

LOCAL FIRST

Medicine records and price snapshots are read from SQLite. Prices are never sourced from model memory.

WEB AS FALLBACK

When enabled, general reference search is domain-limited. It is not a pharmacy scraper or a source of live prices.

03 · BUILD

A small, understandable stack

RUNTIME & LANGUAGE

Bun + TypeScript

Runs the HTTP server, installs project tooling, and executes the TypeScript application.

PERSISTENCE

SQLite

Stores medicine records, snapshot prices, restriction terms, health info, and disclaimer acknowledgement timestamps.

LANGUAGE MODEL

Groq · gpt-oss-120b

Interprets questions and formats tool results. Code and database—not the model—own guard decisions.

REFERENCE SEARCH

Tavily (optional)

Finds general medical-reference excerpts and returns the source links shown to the user.

Vanilla HTML | CSS | Browser JavaScript | JSON HTTP API | Bun test

No separate frontend framework, cloud database, pharmacy partner integration, or deployed service is included in this prototype.

04 · DATA REALITY

What the current dataset can—and can’t—say

15,948

Imported products

Distinct product rows loaded from the 1mg Kaggle CSV snapshot.

11,648

Price entries

Rows with a parseable source price. They are historical snapshot values, not live offers.

0

Verified compositions

The CSV has no ingredient field, so exact-composition alternatives are unavailable.

Available in the CSV

Product name, pack size, rating count, rating, price, and MRP.

Not supplied by this CSV

Verified ingredients, manufacturer, stock, delivery ETA, or capture date.

The source price exceeds the source MRP in many imported records. Both original columns are preserved without guessing what the labels mean.

05 · SAFETY BY DESIGN

Guardrails are code paths—not model promises

INPUT GUARD

Before the LLM

Known restricted terms and selected indirect phrases can return the mandated block response before any model call.

TOOL GUARDS

Before retrieval

Catalogue lookups re-check medicine terms. Search uses a limited domain allow-list and only one call per answer.

OUTPUT GUARD

After generation

Generated answers are checked for known restricted names and common diagnosis phrasing, with limited rewrites.

USER NOTICE

Source & role

Web answers show links and say the information is general—not diagnosis or treatment advice.

Prototype limitation: the restricted-term list currently contains only a small demonstration sample. It is not a complete, official Schedule H list; do not rely on this app for medication decisions.

06 · NEXT MILESTONES

From working prototype to trustworthy data product

01 Verify the Kaggle source license, source URL, and intended use.

02 Acquire an authoritative restricted-drug source and review brand/ingredient mappings.

03 Import MID workbook data only after inspecting columns, overlap, license, and provenance.

04 Validate prices, composition normalization, duplicate handling, and edge-case tests.

05 Obtain an approved pharmacy price feed before claiming live comparisons.

06 Measure false-positive and missed-restricted test cases; document residual risk.

07 Add source dates, license records, audit traces, and operational monitoring.

Data collection rule: no pharmacy scraping until the site's rules and permission allow it.

MEDIBOT · WRAP-UP

Helpful answers start with honest evidence.

A practical prototype today; a clear, testable path toward better data and stronger safety.

Catalogue-backed | Source-aware | Guarded | Not a clinician

Thank you